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Paper 1 of 2 — GPH 314 Note 2 (Seismic Exploration Method)

Level 300 • First Semester • Handout

GPH 314 – SEISMIC EXPLORATION METHODS 1

ELASTIC PROPERTIES OF ROCKS

Theory of Elastic Waves

SEISMIC WAVES are generally referred to as ELASTIC WAVES.

The seismic method utilizes the propagation of these waves through the earth. This propagation depends upon the elastic properties of the rocks. The size and shape of a solid body can be changed by applying forces to the external surface of the body. These external forces are opposed by internal forces which resist the changes in size and shape. This property of resisting changes in size or shape and returning to the undeformed condition when the external forces are removed is called ELASTICITY. A perfectly elastic body recovers completely after being deformed. The theory of elasticity relates the forces which are applied to the surface of a body to the resulting changes in size and shape. The relationship between the applied forces and deformations are described in terms of the concepts of STRESS and STRAIN.

STRESS & STRAIN

The size and shape of a solid body can be changed by applying forces to the external surface of the body. These external forces are opposed by internal forces that resist the changes in size and shape. As a result, the body tends to return to its original condition when the external forces are removed. Similarly, a fluid resists changes in size (volume) but changes in shape. This property of resisting changes in size or shape and of returning to the undeformed condition when the external forces are removed is called elasticity. When external forces are applied to a body, balanced internal forces are set up within it.

STRESS: Stress is a measure of the intensity of these balanced internal forces. It is defined as force per unit area. When a force is applied to a body, the stress is the ratio of the force to the area on which the force is applied. If the force is perpendicular to the area, the stress is said to be a normal stress (or pressure). When the force is tangential to the element of area, the stress is a shearing stress. When the force is neither parallel nor perpendicular to the element of area, it can be resolved into components parallel and perpendicular to the element. Hence, any stress can be resolved into normal and shearing components.

STRAIN: When an elastic body is subjected to stresses, changes in shape and dimensions occur. These changes are called strains. Strain is defined as the relative change (that is, the fractional change) in a dimension or shape of a body. Up to a certain limiting value of stress, known as the yield strength of a material, the strain

is directly proportional to the applied stress (Hooke's Law). This elastic strain is reversible so that removal of stress leads to a removal of strain. If the yield strength is exceeded the strain becomes non-linear and partly irreversible (i.e. permanent strain results), and this is known as plastic or ductile strain. If the stress is increased still further the body fails by fracture. A typical stress-strain curve is illustrated below.

The linear relationship between stress and strain in the elastic field is specified for any material by its various elastic moduli, each of which expresses the ratio of a particular type of stress to the resultant strain.

SEISMIC WAVES

Waves that propagate through the earth as elastic waves are referred to as seismic waves.

There are two broad categories of seismic waves: Body waves and Surface waves.

Body Waves

These are elastic waves that propagate through the Earth's interior. In reflection and refraction prospecting, body waves are the source of information used to image the Earth's interior. Like the ripples on the surface of the pond example described previously, body waves propagate away from the source in all directions. If the speed at which body waves propagate through the Earth's interior is constant, then at any time, these waves form a sphere around the source whose radius is

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dependent on the time elapsed since the source generated the waves. Shown below is a cross section through the earth with body waves radiated from a source (red circle) shown at several different times. In the figure below, ms stands for milli-seconds. One milli-second equals one one-thousandth of a second (i.e., there are one thousand milli-seconds in a second).

Seismic body waves can be further subdivided into two classes of waves: P waves and S waves.

- **P Waves** - P waves are also called primary waves, because they propagate through the medium faster than the other wave types. In P waves, particles constituting the medium are displaced in the same direction that the wave propagates, in this case, the radial direction. Thus, material is being extended and compressed as P waves propagate through the medium. P waves are analogous to sound waves propagating through the air.
- **S Waves** - S waves are sometimes called secondary waves, because they propagate through the medium slower than P waves. In S waves, particles constituting the medium are displaced in a direction that is perpendicular to the direction that the wave is propagating. In this example, as the wave propagates radially, the medium is being deformed along spherical surfaces.

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Most exploration seismic surveys use P waves as their primary source of information.

Surface Waves

Surface waves are waves that propagate along the Earth's surface. Their amplitude at the surface of the Earth can be very large, but this amplitude decays exponentially with depth. Surface waves propagate at speeds that are slower than S waves, are less efficiently generated by buried sources, and have amplitudes that decay with distance from the source more slowly than is observed for body waves.

Like Body waves, there are two classes of Surface waves, Love and Rayleigh waves, that are distinguished by the type of particle motion they impose on the medium.

For our purposes, it is not necessary to detail these differences. Suffice it to say that for virtually all exploration surveys, surface waves are a form of noise that we attempt to suppress. For reflection surveys in particular, suppression of Surface wave energy becomes particularly important, because the amplitudes of Surface waves generated from shallowly buried sources are often observed to be larger than the amplitudes of the Body waves you are attempting to record and interpret. For refraction surveys, Surface waves are less of a problem because we are only interested in the time of arrival of the first wave. Surface waves are never the first arrival. In all of the remaining discussion about seismic waves, we will consider only Body waves.

WAVEFRONTS AND RAYPATHS

In some other geophysical methods, in particular magnetics and resistivity, we often employed two different descriptions of the physical phenomena being observed. For example, when discussing magnetism, we looked at both the strength of the magnetic field and the direction of the magnetic field. When discussing resistivity, we discussed both the electrical potential and current flow.

Similarly, there are two equally useful descriptions of seismic waves: wavefronts and raypaths. The relationship between these two descriptions is shown below.

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- Raypaths - Raypaths are nothing more than lines that show the direction that the seismic wave is propagating. For any given wave, there are an infinite set of raypaths that could be used. In the example shown above, for instance, a valid raypath could be any radial line drawn from the source. We have shown only a few of the possible raypaths.
- Wavefront - Wavefronts connect positions of the seismic wave that are doing the same thing at the same time. In the example shown above, the wavefronts are spherical in shape. One such wavefront would be the sphere drawn through the middle of the dark blue area. This surface would connect all portions of the wave that have the largest possible negative amplitude at some particular time.

In principle and in practice, raypaths are equivalent to the directions of current flow, and wavefronts are equivalent to the equipotential lines described in the resistivity section. They are also equivalent to field direction and strength in magnetism.

Notice that in this example, wavefronts are perpendicular to raypaths. This is in general always true. So, given either a set of wavefronts or a set of raypaths, we can construct the other. This was also true for current flow and equipotential surfaces in resistivity and for field strength and field direction in magnetism.

SEISMIC WAVE SPEEDS AND ROCK PROPERTIES

Before pursuing wave propagation issues any further, let's take a moment to describe how all this wave propagation stuff relates to geologic structure. It's clear from the previous discussion that variations in the speed at which seismic waves propagate through the Earth* can cause variations in seismic waves recorded at the Earth's surface. For example, we've shown that reflected waves can be generated from a planar boundary in seismic wave speed that can be recorded at the Earth's

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surface. How do these velocity variations relate to properties of the rocks or soils through which the waves are propagating?

It can be shown that in homogeneous**, isotropic*** media the velocities of P and S waves through the media are given by the expressions shown below.

Where V_p and V_s are the P and S wave velocities of the medium, ρ is the density of the medium, and μ and k are referred to as the shear and bulk moduli of the media. Taken together, μ and k are also known as elastic parameters. The elastic parameters quantitatively describe the following physical characteristics of the medium.

- **Bulk Modulus** - Is also known as the incompressibility of the medium. Imagine you have a small cube of the material making up the medium and that you subject this cube to pressure by squeezing it on all sides. If the material is not very stiff, you can imagine that it would be possible to squeeze the material in this cube into a smaller cube. The bulk modulus describes the ratio of the pressure applied to the cube to the amount of volume change that the cube undergoes. If k is very large, then the material is very stiff, meaning that it doesn't compress very much even under large pressures. If k is small, then a small pressure can compress the material by large amounts. For example, gases have very small incompressibilities. Solids and liquids have large incompressibilities.
- **Shear Modulus** - The shear modulus describes how difficult it is to deform a cube of the material under an applied shearing force. For example, imagine you have a cube of material firmly cemented to a table top. Now, push on one of the top edges of the material parallel to the table top. If the material has a small shear modulus, you will be able to deform the cube in the direction you are pushing it so that the

cube will take on the shape of a parallelogram. If the material has a large shear modulus, it will take a large force applied in this direction to deform the cube. Gases and fluids cannot support shear forces. That is, they have shear moduli of zero. From the equations given above, notice that this implies that fluids and gases do not allow the propagation of S waves.

Any change in rock or soil property that causes ρ , μ , or k to change will cause seismic wave speed to change. For example, going from an unsaturated soil to a saturated

soil will cause both the density and the bulk modulus to change. The bulk modulus changes because air-filled pores become filled with water. Water is much more difficult to compress than air. In fact, bulk modulus changes dominate this example. Thus, the P wave velocity changes a lot across water table while S wave velocities change very little.

Although this is a single example of how seismic velocities can change in the subsurface, you can imagine many other factors causing changes in velocity (such as changes in lithology, changes in cementation, changes in fluid content, changes in compaction, etc.). Thus, variations in seismic velocities offer the potential of being able to map many different subsurface features.

*Geophysicists refer to the speed at which seismic waves propagate through the Earth as seismic wave velocity. Clearly, in the context of defining how fast seismic energy is transmitted through a medium, speed is a more appropriate word to use than velocity. From our introductory physics classes, recall that velocity implies not only the speed at which something is moving but also the direction in which it is moving (i.e., speed is a scalar quantity, velocity is a vector quantity). Regardless of this well-established difference in the meaning of the two terms, in geophysical jargon, the term velocity is used as a synonym for speed.

**Homogeneous media are those whose properties do not vary with position.

***Isotropic media are those whose properties at any given position do not vary with direction.

SEISMIC VELOCITIES OF EARTH MATERIALS

Most igneous and metamorphic rocks have little or no porosity, and the velocities of seismic waves depend mainly on the elastic properties of the minerals making up the rock material itself. This is also the case with massive limestones, dolomites, and evaporates.

Sandstones, shales, and certain kinds of soft limestones, on the other hand, have more complex microstructures with pore spaces between grains which may contain fluids or softer types of solid material such as clay. For such rocks, velocity is very much dependent on the porosity and on the material filling the pores.

All the types of waves which are important in seismic exploration travel with different velocities in the same type of rock. The P-wave is always the fastest. P-

wave velocities can vary in rocks of the same general type.

The P and S wave velocities of various earth materials are shown below:

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The velocities are generally found to be greater in igneous and crystalline rocks than in sedimentary ones. In the sedimentary rocks they tend to increase with depth of burial and geologic age. Unlike density, there can be a large variation in seismic velocity between different rock types and between saturated and unsaturated soils. Even with this variation, however, there is still considerable overlap in the measured velocities. Hence, a knowledge of seismic velocity alone is not sufficient to determine rock type.

SEISMIC PULSE

When a charge (explosive, air gun, sparker, etc) is detonated the material around it is permanently distorted, a cavity is formed and an outward-travelling seismic impulse (seismic pulse or wavelet) originates.

If the pulse encounters a boundary between two geologic formations with different acoustic impedances (product of the velocity of elastic waves and density of a

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formation), a reflection will occur and a broadened pulse will be received at the Earth's surface after some time. The form and breadth of the pulse and hence its frequency spectrum will depend on the travel time.

AMPLITUDES & ATTENUATION OF SEISMIC PULSE

Seismic waves are reduced in amplitude as they propagate through the Earth. This is influenced by several factors including:

- Geometric Divergence of Energy (Spreading of)
- Absorption of Energy in the medium
- Partial Transmission & Reflection at Acoustic Boundaries
- Subdivision of Energy at an interface (Reflection, Refraction, Conversion)
- Interference with other waves (e.g. Multiple Reflections)
- Influence of measurement system

Of these factors, the first three are the major ones affecting reduction in amplitudes of seismic waves as they propagate. Spreading losses (geometric divergence) are the major cause of the amplitude decrease; absorption is usually the second most important contributor.

Geometric Divergence (Spreading of energy)

When a seismic wave propagates, the energy is spread along the surface of a wavefront. As the wavefront of the seismic energy spreads out from the source with increasing time, the area covered by the wavefront increases. Since the total amount of energy spread over the wavefront remains the same, it must be spread thinly, so to speak, over the wavefront. This means that the amplitude of the seismic impulse

on the wavefront decreases as the wavefront spreads.

For cylindrical and spherical waves, the surface increases with increasing radius.

That is why the energy decreases. The energy of a wave in a given medium is proportional to the square of its amplitude. As a spherical wave spreads out from its source, the energy must be distributed over the area of the sphere, which increases as the square of the sphere's radius.

Absorption of Energy (Inelastic Attenuation)

When seismic energy travels through the earth, it causes particles of material in the earth to vibrate. Each particle rubs slightly against its neighbours, causing friction and consequent loss of seismic energy, which is converted into heat (but only in minute quantities). This form of energy loss causes amplitudes to decrease by a constant factor each second of travel time. In practice, absorption usually causes an amplitude decrease of 10% to 20% each second.

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Partial Transmission Losses

When the seismic pulse passes through the interface between two rock types, some energy is lost from the transmitted pulse due to reflection at the interface. The amplitude, A_T , of the transmitted pulse is:

$$A_T = A_i(1 - RC)$$

Where: A_i = Amplitude of incident pulse

RC = Reflection coefficient for the upper side of the interface.

The transmitted pulse is eventually reflected back from a deeper reflector and on its return path must pass through the same interface again, but in the opposite direction.

The reflection coefficient for the lower side of an interface is the reverse of that for this returning ray:

$$A_T' = A_i'[(1 - (-RC))]$$

$$= A_i'(1 + RC)$$

Thus the 2-way transmission amplitude scalar, T_2 , for the interface is:

$$T_2 = (1 - RC)(1 + RC)$$

$$= (1 - RC^2)$$

This scalar will affect amplitudes of reflections from all deeper interfaces. The scalar value for any one interface will be small, but there are many interfaces in the earth and cumulative loss of amplitude for a deep reflection can be substantial. As an example of reflection coefficient magnitude, 0.1 is a moderately strong reflector, but the corresponding two-way transmission scalar is 0.99, causing a loss of only 1% in amplitude. Typically, the cumulative amplitude loss is 20% to 30% each second, but can be over 50% if there are many strong reflectors in the section.

Paper 2 of 2 — GPH 314 Seismic Exploration Method

Level 300 • First Semester • Handout

GPH 314 – SEISMIC EXPLORATION METHOD I

COURSE OUTLINE

- > Principles of Seismic Method
- > Exploration Objectives & Requirements of Seismic Data Acquisition
- > The Seismic Pulse – Its Generation & Transmission
- > Partition of Seismic Energy at Interface
- > Seismic Energy Reflection, Refraction, Attenuation, & Traveltime-Distance Functions
- > Reflection Time Corrections
- > Field Testing & Procedures
- > Well Velocity
- > The Synthetic Seismogram & the Convolution Model

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Seismic Methods: Refraction and Reflection

Seismic exploration is the use of seismic techniques to map subsurface geologic structure and stratigraphic features with the aim of locating deposits of oil and gas or minerals.

Seismic exploration can also be called Prospecting Seismology, Exploration Seismology and Applied Seismology. The word 'seismic' is derived from the Greek word "seismos" meaning "shock". Seismic has to do with elastic waves. In seismic, the geology is examined using seismic waves.

Seismic methods, as typically applied in exploration seismology, are considered active geophysical methods. In seismic surveying, ground movement caused by some source* is measured at a variety of distances from the source. The type of seismic experiment differs depending on what aspect of the recorded ground motion is used in the subsequent analysis. We do not mean to imply by this statement that any seismic experiment can be done from a given set of observations. On the contrary, the two types of experiments described below have very different acquisition requirements. These acquisition differences, however, arise from the need to record specific parts of the Earth's ground motion over specific distances.

One of the first active seismic experiments was conducted in 1845 by Robert Mallet, considered by many to be the father of instrumental seismology. Mallet measured the time of transmission of seismic waves, probably surface waves, generated by an explosion. To make this measurement, Mallet placed small containers of mercury at various distances from the source of the explosion and noted the time it took for the surface of the mercury to ripple after the explosion. In 1909, Andrija Mohorovicic

used travel-times from earthquake sources to perform a seismic refraction experiment and discovered the existence of the crust-mantle boundary now called the Moho.

The earliest uses of seismic observations for the exploration of oil and mineral resources date back to the 1920s. The seismic refraction technique, described briefly below, was used extensively in Iran to delineate structures that contained oil. The seismic reflection method, now the most commonly used seismic method in the oil industry, was first demonstrated in Oklahoma in 1921. A plaque commemorating this event was erected on the site by the Society of Exploration Geophysicists in 1971.

- **Refraction Seismology** -Refraction experiments are based on the times of arrival of the initial ground movement generated by a source recorded at a variety of distances. Later arriving complications in the recorded ground motion

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are discarded. Thus, the data set derived from refraction experiments consists of a series of times versus distances. These are then interpreted in terms of the depths to subsurface interfaces and the speeds at which motion travels through the subsurface within each layer. These speeds are controlled by a set of physical constants, called elastic parameters that describe the material.

- **Reflection Seismology** - In reflection experiments, analysis is concentrated on energy arriving after the initial ground motion. Specifically, the analysis concentrates on ground movement that has been reflected off of subsurface interfaces. In this sense, reflection seismology is a very sophisticated version of the echo sounding used in submarines, ships, and radar systems. In addition to examining the times of arrival of these, reflection seismic processing extracts information about the subsurface from the amplitude and shape of the ground motion. Subsurface structures can be complex in shape but like the refraction methods, are interpreted in terms of boundaries separating material with differing elastic parameters.

Each of these techniques has specific advantages and disadvantages when compared to each other and when compared to other geophysical techniques. For these reasons, different industries apply these techniques to differing degrees. For example, the oil and gas industries use the seismic reflection technique almost to the exclusion of other geophysical techniques. The environmental and engineering communities use seismic techniques less frequently than other geophysical techniques. When seismic methods are used in these communities, they tend to emphasize the refraction methods over the reflection methods.

*Any of a variety of sources can be used. Typically these sources are manmade, thus satisfying our definition of

an active geophysical survey . One could imagine using natural sources like earthquakes. Experiments that use natural

sources to generate ground motion, however, are considered passive experiments.

Advantages and Disadvantages of Seismic Methods

When compared to the other geophysical methods we've described thus far, the seismic methods have several distinct advantages and several distinct disadvantages.

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If an investigator has deemed that the target of interest will produce a measurable seismic anomaly, you can see from the above list that the primary disadvantages to employing seismic methods over other methods are economically driven. The seismic methods are simply more expensive to undertake than other geophysical methods. Seismic can produce remarkable images of the subsurface, but this comes at a relatively high economic cost. Thus, when selecting the appropriate geophysical survey, one must determine whether the possibly increased resolution of the survey is

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justified in terms of the cost of conducting and interpreting observations from the survey.

Applications of Seismic Method

The seismic method finds diverse applications across various scientific, industrial, and environmental fields. Some of the key applications include:

1. Oil and Gas Exploration:

- Seismic surveys are extensively used in the oil and gas industry to locate potential hydrocarbon reservoirs beneath the Earth's surface.
- The method helps identify subsurface structures, map geological formations, and estimate the size and characteristics of potential reservoirs.

2. Mineral Exploration:

- Seismic methods are employed in mineral exploration to characterize the subsurface and identify potential ore bodies.
- The technique helps in mapping geological structures, determining rock types, and assessing the composition of the Earth's crust.

3. Environmental and Engineering Studies:

- Seismic surveys are used for environmental and engineering applications, including assessing subsurface conditions for construction projects.

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- The method helps evaluate soil properties, identify potential geological hazards, and assess groundwater resources.

4. Civil Engineering and Infrastructure Development:

- Seismic studies are crucial in civil engineering for evaluating the geological conditions of a site before construction.
- The method helps assess the seismic risk of an area, design structures that can withstand earthquakes, and plan infrastructure projects.

5. Geothermal Resource Exploration:

- Seismic methods are employed in the exploration of geothermal resources to identify subsurface structures and assess the potential for geothermal energy extraction.

6. Groundwater Exploration:

- Seismic surveys are used to investigate subsurface conditions and locate potential groundwater aquifers.
- Understanding the geological formations helps in sustainable groundwater management and resource planning.

7. Natural Hazard Assessment:

- Seismic methods play a crucial role in assessing and monitoring natural hazards such as earthquakes, landslides, and volcanic activity.
- The information gathered helps in understanding the subsurface dynamics and potential risks associated with these hazards.

8. Carbon Capture and Storage (CCS):

- Seismic surveys are used in CCS projects to monitor the injection and storage of carbon dioxide in underground reservoirs.
- The method helps ensure the integrity of storage sites and assess the potential for leakage.

9. Archaeological and Cultural Heritage Studies:

- Seismic methods are applied in archaeological studies to non-invasively explore subsurface structures and detect buried archaeological features.
- This can aid in the preservation and documentation of cultural heritage sites.

10. Reservoir Monitoring (4D Seismic):

- In the oil and gas industry, repeated seismic surveys (4D seismic) are conducted to monitor changes in reservoir properties over time.
- This helps optimize production strategies, assess reservoir performance, and identify potential production issues.

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11. Tectonic and Earthquake Research:

- Seismic studies contribute to understanding the Earth's tectonic processes, fault systems, and earthquake mechanisms.
- This information is vital for seismic hazard assessments and earthquake preparedness.

The seismic method's versatility makes it a valuable tool in various disciplines, providing essential insights into the Earth's subsurface for scientific research, resource exploration, and environmental management.

Elastic Waves

When the Earth is rapidly displaced or distorted at some point, the energy imparted into the Earth by the source of the distortion can be transmitted in the form of elastic waves. A wave is a disturbance that propagates through, or on the surface of, a medium.

Elastic waves satisfy this condition and also propagate through the medium without causing permanent deformation of any point in the medium. Elastic waves are fairly common. For example, sound propagates through the air as elastic waves and water waves propagate across the surface of a pond as elastic waves.

In fact, water waves on the surface of a pond offer a convenient analogy for waves propagating through the earth. When a pebble is thrown into a pond, the disturbance caused by the pebble propagates radially outward in all directions. As the ripples move away from their source, notice that there are two distinct ways of looking at the waves as they travel. These two distinct viewpoints are called frames of reference.

- We can view the waves propagating across the surface of the pond from above the pond. At any time, the waves form a circular ring around the source with some radius that is governed by the speed at which the wave propagates through the water and the time elapsed since the wave originated at the source. In this viewpoint, we fix time and we view the wavefield at any location across the entire surface.

- We can view these same waves as they propagate through some fixed location on the surface of the pond. That is, imagine that instead of observing the waves from above the pond, we are in a small boat on the surface of the pond, and we record how the boat moves up and down with respect to time as the wave propagates past the boat. In this viewpoint, we fix our spatial location and view the wavefield at this location at all times.

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These two viewpoints give us two fundamentally different pictures of the exact same wave. Assume that our ripple propagating outward from the source can be approximated by a sine wave.

From the first perspective, we can examine the wave at any location on the surface of the pond at some fixed time. That wave would then be described as shown in the figure below.

In this reference frame, the wave is defined by two parameters: Amplitude and Wavelength.

Amplitude is the maximum displacement that a wave can undergo during a motion. Its unit is metre. It is the peak to trough height of the wave divided by two.

Wavelength is the distance over which the wave goes through one complete cycle (e.g., from one peak to the next, or from one trough to the next).

From our second perspective, we can examine the wave at a fixed location on the surface of the pond as it propagates past us. That is, as time varies. That wave would be described as shown below.

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In this frame of reference, the wave is described by an Amplitude and a Period.

The Amplitude described in this frame is identical to the amplitude described previously.

Period is the time over which the wave is observed to complete a single cycle.

Another commonly used description related to period is the Frequency. Frequency is nothing more than the reciprocal of the period. If the period is measured in seconds (s), frequency has the units of Hertz (Hz), 1/s.

As you might expect, period and wavelength are related. They are related by the speed at which the wave propagates across the surface of the pond, c , where c equals the wavelength divided by the period of the wave.

Wave Representation (Harmonic Wave)

A wave moving sinusoidally with angular velocity, ω and having displacement y whose amplitude is A can be represented by:

$$y = A \sin(\omega t - \phi)$$

$$\text{Or } y = A \sin(\omega t - kx)$$

Where $k = \text{wave number} = 2\pi/\lambda = \text{number of wavelengths in } 2\pi$.

$$\text{Or } y = A \sin 2\pi(t/T - x/\lambda)$$

$$\omega = 2\pi/T = 2\pi f$$

This is for a wave propagating to the right from the origin

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